
testium

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CONTENTS:

1 Overview	1
2 Command Line Interface	3
2.1 -h, --help	3
2.2 -b, --batch-execution	4
2.3 -o, --no-color	4
2.4 -c, --config-file	4
2.5 -r, --run-and-close	4
2.6 -l, --log-file	4
2.7 -d, --define	4
2.8 -p, --report-file	4
2.9 -t, --report-type	4
2.10 -n, --report-pattern	5
2.11 -i, --include-path	5
2.12 -g, --debug	5
3 Modes of operation	7
3.1 Graphical mode	7
3.2 Batch mode	7
3.3 Language server (editor support)	7
3.3.1 Installing the VSCode / VSCodium extension	8
4 TUM file syntax	9
4.1 Configuration files	9
4.1.1 Files loading	10
4.2 Global variables	10
4.2.1 Built-in values	11
4.2.2 Debug mode	11
4.2.3 Test items entries	12
4.2.4 References	12
4.2.5 Dialog values	12
4.2.6 Parameter passing, variable expansion and evaluation	12
4.3 Test Items	13
4.3.1 Items common attributes	13
4.3.2 Container items GUI default folding	17
4.3.3 check test item	17
4.3.4 console test item	17
4.3.5 dialog_choices test item	20
4.3.6 dialog_image test item	22
4.3.7 dialog_message test item	23
4.3.8 dialog_note test items	23
4.3.9 dialog_question test item	23
4.3.10 dialog_references test item	24
4.3.11 dialog_value test items	25



4.3.12	git test item	25
4.3.13	group test item	26
4.3.14	jsonrpc test item	26
4.3.15	let test item	29
4.3.16	loop test items	30
4.3.17	lua_func test item	31
4.3.18	parallel test item	33
4.3.19	plot test item	35
4.3.20	py_func test item	38
4.3.21	pytest test item	41
4.3.22	report test item	41
4.3.23	run test item	42
4.3.24	sleep test item	43
4.3.25	unittest test item	43
4.4	Includes	45
4.5	Templates	45
4.5.1	In the main test file	45
4.5.2	In <code>!include</code> directive	46
4.6	Reports	46
4.6.1	Declaring exports	46
4.6.2	Export attributes	47
4.6.3	Built-in formats	48
4.6.4	command export — external tool	49
4.6.5	Custom export formats (plugins)	49
4.6.6	Report database schema	51
4.7	Test outputs	51
4.8	Post execution	51
4.9	Sub-sequence references	52
4.10	Test documentation	52
4.10.1	Unittest	53
5	Python helper library	55
5.1	Global variables helper functions	55
5.2	Plot helper functions	56
5.3	Other helper functions	56
5.4	Debug mode	56



OVERVIEW

testium is an automated test framework written in Python by François Dausseur. It uses the Qt6 graphical framework.

It has been developed since 2013 with production and development testing in mind.

Its function is to automate the execution of tests. It can be invoked either as command line application or as a graphical interface application.

The tool also generates test reports and lets you customize them.

Its main features are:

- YAML test description,
- Test configuration files in YAML,
- Full range of pre-existing Test items,
- Test steps, loops,
- Dynamic variables expansion at test runtime,
- Conditional test step execution,
- Modularity of tests (reusable test sequences),
- etc.

These features let the test engineer write efficient and robust tests.

Each test is described with the help of a [YAML](https://yaml.org/)¹ file having .tum as extension. This file is analyzed and then displayed as a tree in the GUI (see Figure above).

¹ <https://yaml.org/>

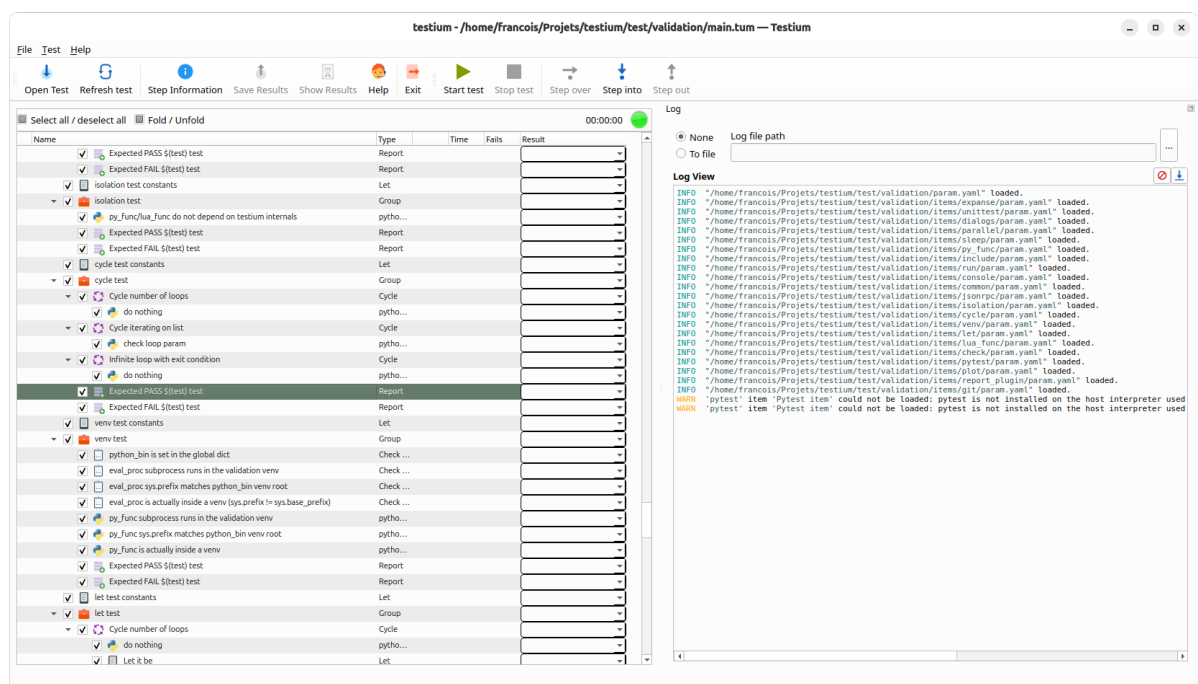


Fig. 1.1: testium

COMMAND LINE INTERFACE

```
usage: testium.pyw [-h] [--version] [-b] [-c CONFIG_FILE [CONFIG_FILE ...
→]] [-r] [-l LOG_FILE]
                        [-d DEFINE [DEFINE ...]] [-p REPORT_FILE] [-t
→{sqlite,json,junit,html,text}]
                        [-n REPORT_PATTERN [REPORT_PATTERN ...]] [-i
→INCLUDE_PATH [INCLUDE_PATH ...]] [-o] [-g]
                        [test_file]

positional arguments:
test_file                the test script file

optional arguments:
-h, --help                show this help message and exit
--version                Returns the version of testium
-b, --batch-execution    Executes the test in batch mode
-c CONFIG_FILE [CONFIG_FILE ...], --config-file CONFIG_FILE [CONFIG_FILE ..
→.]
-o, --no-color            Deactivates stdout colors in batch mode
-r, --run-and-close       Runs the test then closes the application
-l LOG_FILE, --log-file LOG_FILE
                        log file name
-d DEFINE [DEFINE ...], --define DEFINE [DEFINE ...]
                        Configuration passed to the executed tests.
-p REPORT_FILE, --report-file REPORT_FILE
                        report file name
-t {sqlite,json,junit,html,text}, --report-type
→{sqlite,json,junit,html,text}
                        report file type
-n REPORT_PATTERN [REPORT_PATTERN ...], --report-pattern REPORT_PATTERN
→[REPORT_PATTERN ...]
                        report file pattern
-i INCLUDE_PATH [INCLUDE_PATH ...], --include-path INCLUDE_PATH
→[INCLUDE_PATH ...]
                        Python modules search path
-g, --debug                GUI debug mode
```

2.1 -h, --help

Prints the usage message shown above.



2.2 -b, --batch-execution

Executes the test in text mode. Qt does not need to be installed in this mode.

2.3 -o, --no-color

Disables colored output in batch mode.

2.4 -c, --config-file

This option provides configuration file(s) from the command line. The configuration files format and content is detailed in the [config files](#) section.

If this parameter is not given while calling *testium*, the default configuration files will be used.

2.5 -r, --run-and-close

This parameter makes testium close immediately after running the `test_file` passed on the command line.

If there is no `test_file` argument passed, this option is ignored.

2.6 -l, --log-file

Path of the log file for the test execution. If not provided, the log is written to a temporary folder.

2.7 -d, --define

Defines one or more variables in the form `VARIABLE1=value1 VARIABLE2=value2 ...`. Then, these variables are available from the test scripts, using the [global variables testium](#) feature.

2.8 -p, --report-file

Path of the report file, stored during the test execution.

This option is only useful in [batch mode](#).

2.9 -t, --report-type

This option is used together with option `-p` and defines the type of report to generate.

Please read the [reports](#) section for more details on the possible types of report.

2.10 -n, --report-pattern

This option is used in conjunction with option *-p* and defines the report pattern(s) used to filter the report results which will be included in the report file.

More details in *reports* section.

2.11 -i, --include-path

Additional Python paths. These paths are appended to the `sys.path`² python variable.

2.12 -g, --debug

This option is only useful when debugging *testium* in vscode in *graphical mode*.

² <https://docs.python.org/3/library/sys.html?highlight=sys%20path#sys.path/>

MODES OF OPERATION

3.1 Graphical mode

testium was originally designed as a graphical (GUI) application.
Run the `testium` command to start it. This is the default mode.

3.2 Batch mode

Batch mode executes a test in text mode. In this mode, the test does not start any graphical interface.

Listing 3.1: call a test in batch mode

```
testium -b test/my_test/main.tum
```

3.3 Language server (editor support)

testium ships a [Language Server Protocol](https://microsoft.github.io/language-server-protocol/)³ server so that `.tum` files get editor assistance — completion of test item types, hover documentation of their parameters, and an outline view — in any LSP-capable editor.

The server speaks LSP over standard input/output and is started with:

Listing 3.2: start the language server

```
testium lsp
```

It is not meant to be launched directly by the user: an editor's LSP client spawns it and drives the exchange. A VSCode / VSCode client extension, *testium_assist*, is provided for that purpose; any other LSP client (Neovim, Emacs `lsp-mode`, ...) can be pointed at *testium lsp* as well.

The information the server exposes is the test item schema, which can also be dumped as JSON for inspection or tooling:

³ <https://microsoft.github.io/language-server-protocol/>

Listing 3.3: dump the item / parameter schema

```
testium schema
```

Because the schema is built from *testium* itself, every new item type or parameter becomes available in the editor on the next *testium* upgrade, with no change to the client.

The language server is included in the pre-built binary, Flatpak and Appliance releases. For a source or wheel installation, pull the optional `lsp` dependencies:

Listing 3.4: enable the language server for a wheel / source install

```
pip install 'testium[lsp]'
```

3.3.1 Installing the VSCode / VSCodium extension

The *testium_assist* client extension is published on [Open VSX](https://open-vsx.org/extension/testium/testium-assist)⁴, the registry used by VS-Codium, Cursor, Windsurf, Eclipse Theia and code-server. In those editors, open the Extensions view and search *testium-assist*, or install it from the command line:

Listing 3.5: install in VSCodium and other Open VSX editors

```
codium --install-extension testium.testium-assist
```

Microsoft *VSCode* uses a different marketplace that does not list Open VSX extensions, so install the packaged `.vsix` by hand. Download it from the Open VSX page linked above, then either choose *Extensions* → ... → *Install from VSIX...* in the UI, or run:

Listing 3.6: install the `.vsix` in Microsoft VSCode

```
code --install-extension testium-assist-0.1.0.vsix
```

The extension launches *testium lsp*, so the *testium* command must be on the `PATH`. If *testium* is installed elsewhere — a specific binary or an Appliance — point the *testium.serverPath* setting at it instead.

Once installed, open a `.tum` file: completion of item types, hover documentation and the outline view become available. If nothing happens, check that no `files.associations` entry forces `*.tum` to another language (it must stay the `tum` language the extension provides).

⁴ <https://open-vsx.org/extension/testium/testium-assist>

TUM FILE SYNTAX

testium is a python-based tool which uses a YAML based description file to operate tests: the TUM file.

The description of tests is based on the definition of test sequences. There is a main YAML element which is the *testium* tool entry point.

All other steps are listed in the step list of the main. Some steps, such as loop or console items, are themselves lists of steps.

In YAML, nesting is expressed by indentation.

YAML language auto detects data type so that it is not necessary to cast the element type explicitly. See [YAML home page](https://yaml.org/)⁵ for further information on the language.

The example below shows a basic implementation of the TUM description file:

Listing 4.1: main test file

```
main:
  name: Test example
  steps:
    - test_item1:
      name: test_1
    - loop :
      name: test cycle
      iterator: 5
      steps:
        - test_item:
          name: test_2
        - test_item:
          name: test_3
```

4.1 Configuration files

A configuration file can be specified in the *.tum* file or by the command line. This configuration file is optional and must be a YAML file.

The configuration files must have the *.yaml* or *.yml* file name extension.

During the test script loading process, the values defined in these configuration files are added to the global variables and are then accessible from the test items and scripts (cf. [global variables](#)).

The parameter file can be specified in the *.tum* file root:

⁵ <https://yaml.org/>

Listing 4.2: configuration files definition in the main *.tum* test file

```
config_file:
  - config1.yaml
  - config2.yaml

main:
  name: Test example
  [ ... ]
```

Listing 4.3: example of configuration file: *param.yaml*

```
parameter1: value1
parameter2: 1234
parameter3: <| 12.34 * 2 |>
parameter4:
  - $(parameter1)
  - $(parameter3)
parameter5:
  sub_param1: sub_value1
  sub_param2: $(parameter4)
```

If nothing is specified, the *param.yaml* is automatically loaded, if present in the test directory.

4.1.1 Files loading

The YAML configuration files variables are evaluated directly and accessible from TUM tests description files and also from *python* and *lua* function test items.

See more details [below](#).

4.2 Global variables

The global variables feature lets test items and test scripts access a shared, global variables database.

The global variables dataset is populated from various sources:

- the *command line*,
- *built-in values*,
- the *configuration files*,
- some test items results,
- the *helper library API*, accessible from python scripts.

These global variables are used for variable expansion in scripts (see [variables expansion](#)).

Another possible usage of the global variables is to share persistent data between test steps.

A library allowing python functions to access global variables is available from the python scripts. See details in section [helper library](#).

Besides the values from the default *param.yaml* or defined configuration files, the global variables also contain

- built-in specific value (see [below](#)),

- values returned by test items.

4.2.1 Built-in values

The following keys are automatically accessible through the testium library API (see [helper library](#))

- `test_directory`: the absolute path of the directory of the main .tum file,
- `test_main_file`: the main .tum file,
- `os`: the name of the platform which is used. Can be Linux or Windows,
- `host_name`: The name of the host on which testium is running,
- `home`: home directory of the current user,
- `testrun_date`: The date when the test has started (as a string) in format YYYY-MM-DD,
- `testrun_time`: The time when the test has started (as a string) in format HH:MM:SS,
- `test_name`: The name of the file being executed without extension,
- `home`: the path of the current user's home directory,
- `test_outputs`: list of the paths of the test log and test report (if any),
- `last_step_result`: test result of the last step (see [Items common attributes](#)),
- `ts_start_<item_name>`: timestamp at the start of test item execution (see [Items common attributes](#)),
- `ts_end_<item_name>`: timestamp at the end of test item execution (see [Items common attributes](#)),
- `ts_duration_<item_name>`: duration of test item execution in seconds (see [Items common attributes](#)),
- `cn_<test_name>`: console test item result (see section [console test item](#)),
- `pfn_<func_name>`: py_func test item result (see section [py_func test item](#)),
- `lfn_<func_name>`: lua_func test item result (see section [lua_func test item](#)),
- `cs_<test_name>`: dialog_choices test item result (see section [dialog_choices test item](#)),
- `loop_param`: loop iterator (available from within a loop item, see [loop test items](#)),
- `loop_index`: loop index (available from within a loop item, see [loop test items](#)),
- `loop_count`: loop number (available from within a loop item, see [loop test items](#)). If the loop is infinite, its value is the Python constant `inf`.

4.2.2 Debug mode

Debug mode can be enabled by defining the global variable `test_debug`.

For example, it can be defined in the configuration file as:

Listing 4.4: example of configuration file: param.yaml

```
[...]  
test_debug: True  
[...]
```

It can also be defined from the command line with the option `-d test_debug`.

When debug mode is enabled, additional information is displayed in the log window.

Some *helper library functions* are available to query the state of the debug mode.

4.2.3 Test items entries

All test items attributes can be global variable entries; when a value is written as `$(<name>)`, the name is looked up in the global variables dataset.

4.2.4 References

If the `dialog_references` test item has been included (see *dialog_reference test item*), the global dict will contain the result of this test item in the key `tested_items`.

4.2.5 Dialog values

All dialog returned values are inserted in the global variables entries with the key value being the test item name attribute (see *dialog_value test item*).

4.2.6 Parameter passing, variable expansion and evaluation

One of the most useful functionalities for scalability and flexibility of the .tum files is the ability to expand variables at test runtime.

It is done by replacing any occurrence of `$(my_global)` with the content of the variable in the global variables entries (see *global variables*).

The variable substitution is recursive and checks all the occurrences of the `$(x)` pattern in a string.

It is also possible to evaluate embedded Python expressions when parameters are passed. It is done by using the `<| expr |>` pattern in a string. `expr` must be a valid Python expression.

Below are illustrated simple and more complicated cases of expansion and evaluation depending on their pattern.

Listing 4.5: variables expansion and evaluation

```
- let:  
  name: Dynamic variables expansion  
  key: $(test)_PASS  
  values:  
    - expanse_select: <|"$(expanse_select)".replace("o", "a")|>  
    - expanse_index: $(expanse_index_$(expanse_select))  
    - expanse_table: $(expanse_table_$(expanse_select))  
    - expanse_eval: <|$(expanse_index) == 1|>
```


4.3 Test Items

All *testium* steps are described in sequence as test items in the step list of the main test item (and possibly of a loop test item).

TUM file main item is itself a variant of test item with name and step-list attributes.

Note

Each test item declares the parameters it accepts. When a .tum file uses a key the item does not know, *testium* emits a warning listing the accepted parameter names (catching typos such as `param_filee` for `param_file`); a missing **required** parameter aborts loading with an error pointing at the source .tum file. Valid existing tests are unaffected.

4.3.1 Items common attributes

All test item types share the common attributes listed in the following table. They are all optional; the table also gives their default values.

Table 4.1: test items common attributes

Parameter name	Default value	Description
name	test item type	This is the test item name as displayed in the test tree window of the testium. This attribute is also supported by actions of console, jsonrpc or plot test items. Default value, if not provided, is the test item type.
stop_on_failure	False	If stop_on_failure is set to True, the test sequence execution stops on test item failure and no further test items are executed, except those with the execute_on_stop attribute set (see below) Not all test item types honor this attribute. For example it makes sense to use it for unittest test type because it can contain many sub-tests, but not for sleep test type. In cycles, it means that the child sequence execution is stopped at first failure. It also means that the remaining loops are not executed.
execute_on_stop	False	When this attribute is set to True, the test item is always run, even on test failure of any test before. This feature is useful, to end the test sequence properly on test failure (switch off power supplies, climatic chamber temperature set to ambient temperature....)
skipped	False	The test item execution is to be skipped during test sequence execution if set to True. It will be displayed as failed in the report.
no_fail	False	The result of the test step is forced to PASS if this attribute is set to true.
doc	""	Documentation for the test item that appears in the test doc field and the contextual text window in the testium GUI.
Key	/	This attribute defines a key which will be attached to the test result and which allows results to be filtered during the report generation.
report	/	This attribute defines values (a dictionary) which will be added in the data field of the report.
condition	/	The test item is executed if its condition attribute content is a boolean True. see Conditional execution .
process_result	/	Process an evaluation of the process_result and store it in the result see Process result for details.
expected_result	/	Expected result value or string. see Expected result for details.
store_result	/	Store the test result in a global variable. see Store result for details.

last test result

The global variable `last_step_result` is automatically set at the end of a test item execution.

If the corresponding test item does not return any actual value, the content of the `last_step_result` variable will be the test success (PASS, FAIL or SKIP).

If the test item returns a value, the `last_step_result` variable will contain the returned value.

The main test items returning a value are:

- *console* test item,
- *jsonrpc* test item,
- *dialog references* test item,
- *dialog value* test item.

Test timings

After the execution of a test step, the following global variables are set :

- `ts_start_<item_name>`
- `ts_end_<item_name>`

and

- `duration: ts_duration_<item_name>`

See *global variables* for more detail on how to access to global variables from test items and scripts.

Skipped test items

A variable named `skipped_test_item` can be defined in the global variable entries or in configuration file (see *config files*) as a list of item to be skipped.

Conditional execution

The condition attribute content must be a boolean value (if not True, the condition is considered not met).

Process result

The `process_result` attribute can be applied to all the test items. However, its behavior differs depending on whether the test item returns a value.

The `process_result` attribute content is evaluated as a python line.

The special `$(result)` variable is replaced in the `process_result` attribute content with the test result value.

`process_result` is evaluated before `expected_result`.

If the result of the evaluation is a boolean, the test will be *PASSED* if True, and *FAIL* otherwise.

Expected result

The `expected_result` attribute can be applied to all the test items. However, its behavior differs depending on whether the test item returns a value.

The test items returning a value are:

- *dialog_references test item*
- *dialog_value test items*
- *py_func test item*
- *dialog_choices test item*
- *json_rpc test item*

For test items which don't return a value, the `expected_result` attribute content is compared to `PASS` or `FAIL`.

The `expected_result` attribute content is a simple comparison with `$(result)`.

The test is *PASS* if the result equals `expected_result`, and *FAIL* otherwise.

The special `$(result)` variable is replaced in the `expected_result` attribute content with the test result value.

Store result

The `store_result` attribute stores the test result into a named global variable, making it available to subsequent test items via `$(variable_name)`.

If the test item returns a value (e.g. `py_func`, `json_rpc`), that value is stored. If `process_result` is also specified, the stored value is the post-processed result.

If the test item produces no value (result is `None`), the stored value is the test status string: `"PASS"` or `"FAIL"`, evaluated after `expected_result` but **before** `no_fail`. This ensures the real outcome is captured even when `no_fail: True` would otherwise mask a failure.

Listing 4.6: Store a function return value

```
- py_func:
  name: Read sensor
  func_name: read_temperature
  store_result: temperature

- py_func:
  name: Check temperature in range
  func_name: check_range
  param: [$(temperature), 20, 30]
```

Listing 4.7: Store a post-processed value

```
- py_func:
  name: Get firmware version string
  func_name: get_version
  process_result: '$(result)'.split('.')[0]'
  store_result: fw_major
```

Listing 4.8: Store the pass/fail status of a test with no return value

```
- console:
  name: Send command
  console_name: device
  steps:
    - writeln: reboot
    - read_until: {expected: "ready", timeout: 10}
  store_result: reboot_status

- py_func:
  name: Use reboot status
  func_name: log_status
  param: [$(reboot_status)]
```

key and report attributes

Listing 4.9: Example of key and report common attributes usage

```
- check:
  name: Example of result specific to the step 001
  values:
    - $(last_step_result) == PASS
  key:
    - GID-1510554_step_1
  report:
    reported_list: <| random.sample(range(0,20), k=10) |>
    reported_float: <| math.sqrt(float(1)) |>
    reported_str: This is my reported sentence
```

4.3.2 Container items GUI default folding

The container items are items which are the parent of other test items. For example loops and groups are container test items.

In the GUI, to make a container test item folded when the test is opened, place a . character before the test item declaration.

See an example below:

Listing 4.10: example of loop folded by default in the GUI

```
- .loop:
  doc: An example loop
  name: An example loop
  ...
```

4.3.3 check test item

The check test item returns the result of a python string evaluation:

Listing 4.11: example of check test item usage

```
- check:
  name: check test item example
  values:
    - '"tictactoe" in "$(my_global_string)'"
```

4.3.4 console test item

The console test item is of the form:

Listing 4.12: example of console test item usage

```
- console:
  name: test name in GUI
  console_name: console name in dict
  steps:
    - open:
      protocol: telnet
      telnet_host: $(target_ip)
```

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```
    telnet_port: $(target_port)
-   writeln: reset
-   read_until: {expected: U-Boot, timeout: 50}
-   write: $(boot_vxworks_1)
-   writeln: $(boot_vxworks_2)
-   read_until:
      expected: U-Boot
      timeout: 15
-   read_until:
      expected: Something that will never occur
      timeout: 5
      no_fail: True
      mute: True
-   close:
```

Attributes

Besides the common test item attributes, the console test item has specific attributes:

- `console_name`: console instance name
- `write_delay`: optional parameter giving the delay to wait in milliseconds between each character sent.
- `steps`: a sequence of actions to be applied to the console, as listed above.

The console test item steps accept the parameters and configurations defined in the next sections.

All the following actions support the name attribute. The name is concatenated with the step type in the *testium* GUI, and repeated in the test log and reports.

open action

The open action initializes the console with the attributes defined as described below. The console instance is then added to the `console_instances` entry of the global variables (cf [global variables](#)).

Open step accepts the following attribute:

- `protocol`: Setting of the console protocol; supported protocols are listed in the table below
- The other attributes depend on the protocol in use and are listed in the table below

Table 4.2: console protocols

Protocol	protocol parameter	Description
telnet	telnet_host	hostname of the target.
	telnet_port	port of the telnet server of the target.
ssh	ssh_host	Hostname or IP address of the target.
	ssh_user	User name for the SSH connection.
	ssh_pwd	Password (optional).
serial	serial_port	Serial port to the target.
	serial_baudrate	Baud rate of the serial connection.
	buffered	Optional boolean parameter. If False, it forces the console to read the device directly. Default: True.
rawtcp	tcp_host	hostname of the target.
	tcp_port	port of the rawtcp server of the target.
terminal	terminal_path	Path of the terminal console.
	shell	Shell to execute in the terminal Default: /usr/bin/env bash

- log: is available only for Telnet and Serial console and is a path to a folder or a file, where the log will be stored.

close action

The close action closes the console devices and removes its instance from the console_instances list accessible in the global variables (cf [global variables](#)).

No parameters required for this action.

write action

write action takes as parameter the string to be written on the console.

writeln action

writeln function is similar to the write function except that a 'n' (newline) character is sent at the end of the string to be written.

read_until action

The read_until action waits for a string pattern from the console. Its parameters are listed below:

- expected: the pattern(s) to wait for. It accepts either a **single value** or a **list of values**; when a list is given the action succeeds as soon as **any** of the values is seen.
- regex: Boolean value (True or False, default False). When True every expected entry is interpreted as a Python regular expression (searched in the incoming stream, not anchored) instead of a literal string.
- timeout: Timeout setting for the action (in seconds)
- no_fail: Boolean value (True or False). If True, no error is reported when the expected input is not read
- mute: Boolean value (True or False). If True, the data read is not logged

Listing 4.13: matching several values, and with a regular expression

```
# succeeds as soon as one of the three strings is received
- read_until:
  expected: [login:, "Password:", "$ "]
  timeout: 10

# regex: wait for "version X.Y.Z" with any numbers
- read_until:
  expected: 'version \d+\.\d+\.\d+'
  regex: True
  timeout: 5
```

The text read by the `read_until` action is stored in the global variable named `cn_<test_name>` (See [global variables](#) for more detail on accessing global variables from test items and scripts). When a list of values is given, the report also records, under the matched key, which pattern actually matched.

Note

regex matching scans a bounded tail of the received stream (`Console.REGEX_WINDOW` characters), so a pattern that could only match after a very large amount of output — or across more than that window — may not be detected. Literal matching (the default) has no such limit.

In the example above, the global variable `$(cn_test name in GUI)` would be created at the end of the step. It contains the data that was read.

4.3.5 dialog_choices test item

This test item displays a dialog asking a question and waits for the user to select entries from a defined list of items.

These selectable items can be passed as a tree.

The [Figure 4.1](#) displays an example of this item.

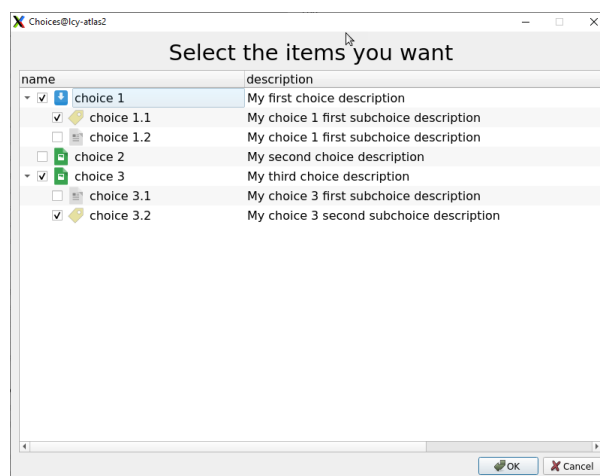


Fig. 4.1: choices dialog

The item parameters corresponding to [Figure 4.1](#) are shown below.

Listing 4.14: example of choices_dialog test item usage

```

- dialog_choices:
  name: Choices
  question: Select the items you want
  icon: $(test_directory)/document.png
  choices:
    - name: choice 1
      description: My first choice description
      icon: $(test_directory)/document-save.png
      choices:
        - name: choice 1.1
          description: My choice 1 first subchoice description
          icon: $(test_directory)/Label.png

        - name: choice 1.2
          description: My choice 1 first subchoice description

    - name: choice 2
      description: My second choice description
      icon: $(test_directory)/image.png

    - name: choice 3
      description: My third choice description
      icon: $(test_directory)/image.png
      choices:
        - name: choice 3.1
          description: My choice 3 first subchoice description

        - name: choice 3.2
          description: My choice 3 second subchoice description
          icon: $(test_directory)/Label.png

```

Attributes

The supported attributes of the dialog_choices test item are:

- **question:** Question to be displayed in the dialog box.
- **choices:** List of the choices presented to the user.
- **icon:** Optional. Path of the icon used in the selection tree, for all the items by default.
- **auto_result:** Optional. Selection used in batch / non-interactive mode (choice path or label). If not set, the step fails.

Choices attribute content

Each choice element is a dictionary which can have the following attributes

- **name:** name of the choice.
- **description:** description of the choice.
- **icon:** Optional. Path of the icon displayed in the selection tree in front of the corresponding choice.
- **choices:** List of sub-choices presented to the user (recursive).

Feature

The `dialog_choices` test item creates the `cs_<name of test item>` entry in the global dictionary.

In the example above, the global variable name containing the result of the test item would be `cs_Choices`, and it would contain an object of this form:

Listing 4.15: example of result of the `dialog_choices` test item

```
[
  {
    'name': 'choice 1',
    'checked': True,
    'choices': [
      {
        'name': 'choice 1.1',
        'checked': True
      },
      {
        'name': 'choice 1.2',
        'checked': False
      }
    ]
  },
  {
    'name': 'choice 2',
    'checked': False
  },
  {
    'name': 'choice 3',
    'checked': True,
    'choices': [
      {
        'name': 'choice 3.1',
        'checked': False
      },
      {
        'name': 'choice 3.2',
        'checked': True
      }
    ]
  }
]
```

See [global variables](#) for more detail on how to access global variables from test items and scripts.

4.3.6 dialog_image test item

This test item displays an image within a dialog box.

`dialog_image` test item has the following description format

Listing 4.16: example of `dialog_image` test item usage

```
- dialog_image:
  name: dialog image test item
  question: operator question
  filename: imageToBeDisplayed.jpg
```

Attributes

`dialog_image` has the following specific attributes:

- `question`: Question to be displayed in the dialog box
- `filename`: File name of the image to be displayed in the dialog box.
- `auto_result`: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails.

Feature

The test returns a FAIL if the answer is No and PASS if yes.

4.3.7 dialog_message test item

This test item displays a simple dialog showing a message. dialog_message test item has the following description format

Listing 4.17: example of dialog_message test item usage

```
- dialog_message:  
  name: dialog value test item  
  question: operator question
```

Attributes

dialog_message has the following specific attribute:

- question: Sentence to be displayed in the dialog box
- auto_result: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails.

Feature

Displays the message; no value is returned.

4.3.8 dialog_note test items

This test item displays a simple dialog that lets the operator enter text, and prints the entered text in the logs.

dialog_note test item has the following description format

Listing 4.18: example of dialog_note test item usage

```
- dialog_note:  
  name: dialog value test item  
  question: operator question
```

Attributes

dialog_note has the following specific attribute:

- question: Question to be displayed in the dialog box
- auto_result: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails; cancel marks it cancelled; any other value makes it succeed with auto_value.
- auto_value: Optional. Note text used in batch mode when auto_result is set.

Feature

Prints the entered text in the log.

4.3.9 dialog_question test item

This test item displays a dialog asking a yes/no question; the answer determines the test result.

dialog_question test item has the following description format

Listing 4.19: example of dialog_question test item usage

```
- dialog_question:  
  name: dialog value test item  
  question: operator question
```

Attributes

dialog_question has the following specific attribute:

- question: Question to be asked in the dialog box
- auto_result: Optional. Answer used in batch / non-interactive mode ('yes' or 'no'). If not set, the step fails.

Feature

The test returns a FAIL if the answer is No and PASS if yes.

4.3.10 dialog_references test item

This test item displays a dialog asking a question and waiting for references of the devices under test.

This test item has the following format:

Listing 4.20: example of dialog_references test item usage

```
- dialog_references:  
  name: ask for a reference  
  question: Please give the reference of the product  
  reference:  
    - ref 1  
    - ref 2/rev
```

The example above displays the dialog box in [Figure 4.2](#).

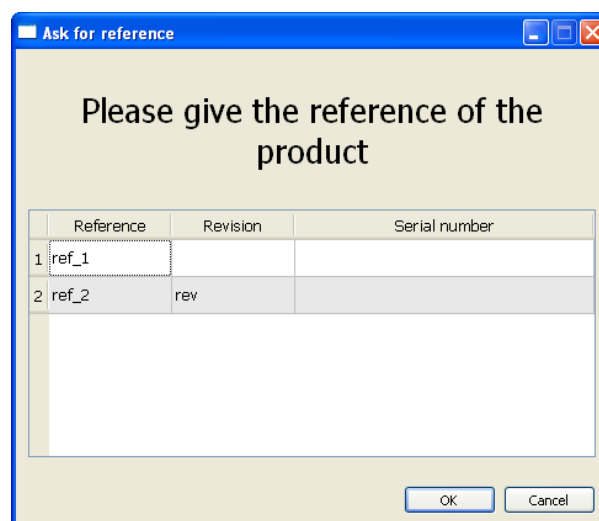


Fig. 4.2: dialog reference

Attributes

question and reference are mandatory.

- **question:** Question to be displayed in the dialog box.
- **reference:** Each entry in this list corresponds to one row to fill in the dialog.
- **auto_result:** Optional. Outcome used in batch / non-interactive mode. If set, the step succeeds with the pre-filled references; if not set, it fails.

Every field for a reference can be pre-filled by separating each field with a '/' (cf [Figure 4.2](#)).

Feature

The dialog references test item creates the `tested_items` entry in the `global_dict` global variable. This entry is a list of dictionaries of this form:

Listing 4.21: example of `tested_items` global variable result of `dialog_reference` test item

```
[{'reference': 'XXXXX', 'revision': 'YYYYY', 'serial': 'ZZZZZ'}, ...]
```

4.3.11 dialog_value test items

This test item displays a simple dialog asking a question and returning the entered value.

`dialog_value` test item has the following description format

Listing 4.22: example of `dialog_value` test item usage

```
- dialog_value:
  name: dialog value test item
  question: operator question
```

Attributes

`dialog_value` has the following specific attribute:

- **question:** Question to be displayed in the dialog box
- **default:** default value to place in the dialog form (optional)
- **auto_result:** Optional. Outcome used in batch / non-interactive mode. If not set, the step fails; `cancel` marks it cancelled; any other value makes it succeed with `auto_value`.
- **auto_value:** Optional. Value used in batch mode when `auto_result` is set.

Feature

The entered value is stored in the global variables under the `dialog_value` test item name as key.

4.3.12 git test item

The `git` test item records the state of one or more git repositories. It has the following description format

Listing 4.23: git test item usage example

```
- git:
  name: git test item
  repo: [$(test_directory), "/path_to/another/repo"]
```

Attributes

- repo: a path, or list of paths, to the root of the git repository or repositories to track.

4.3.13 group test item

This element is of the following form:

Listing 4.24: group test item usage example

```
- group:
  name: Group Item
  condition: <| "$(OS)" == "Linux" |>
  steps:
    - unittest:
      test_file: test_prod_alpha_13.py
      test_method:
        ...
    - sleep:
      timeout: 10
```

The group element is used to run a sequence of items as a group.

Attributes

- The steps list describes the sequence executed in the group. It is a list of any *testium* test items.

4.3.14 jsonrpc test item

The *jsonrpc* test item is used to access jsonrpc servers, by sending queries and analysing the answers. It supports JSONRPC [v1.0⁶](https://www.jsonrpc.org/specification_v1) or [v2.0⁷](https://www.jsonrpc.org/specification).

This test item can access the jsonrpc server by using an existing *console* or directly using a UDP protocol. Two low level *adapters* can then be chosen: *udp* or *console*.

Example of jsonrpc test item with the console adapter:

Listing 4.25: json_rpc test item usage example

```
- json_rpc:
  name: JSONRPC console Query
  doc: JSONRPC console Query not waiting (only send)
  console:
    name : jsonrpc_server
    prompt: "@@>"
  timeout: 1
  version: "2.0"
  steps:
```

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⁶ https://www.jsonrpc.org/specification_v1

⁷ <https://www.jsonrpc.org/specification>

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```

- query:
  method: echo
  params:
    a: Hello world
    b: [0, 1, 2, 3]
  id: 3095372
  no_wait: True

- [...]

- json_rpc:
  name: JSONRPC console Reception
  doc: JSONRPC console reception of the previous request
  console: {name : jsonrpc_server}
  timeout: 1
  steps:
    - receive:
      name: console reception
      id: 3095372
      timeout: 0.5

```

Attributes

the jsonrpc attributes are:

- `timeout`: global communication timeout in seconds. It is a floating point number.
- `version`: "1.0" or "2.0" (as a string) depending on the version of the JSONRPC standard which is supported.
- `mute`: a boolean giving the verbosity of the jsonrpc exchanges on the log output.
- An *Adapter* is to be chosen between:
 - Console,
 - UDP,
- `steps`: a sequence of actions as described in the sections below.

Steps

A jsonrpc step can be one of the following:

- `open`: used by the UDP adapter to open the socket explicitly,
- `close`: used by the UDP adapter to close the socket explicitly,
- `query`: performs a complete or partial JSONRPC call,
- `receive`: used to receive the JSONRPC result of call previously done by the query action.

If no `expected_value` attribute is defined for query or receive actions, the success of the step will depend on the value returned by the JSONRPC frame. The protocol itself defines a way to report whether the remote procedure succeeded or failed.

All the actions support the `name` attribute. The name is concatenated with the action type in the *testium* GUI, and repeated in the test log and reports.

adapter attributes

The adapters attributes are listed in the table below.

Table 4.3: jsonrpc adapters

adapter	attribute	type	Description
Console	console	<i>dictionary</i>	The console adapter configuration
	console.name	<i>string</i>	The name of the console which will be retrieved from the global variables . See also the console test item .
UDP	console.prompt	<i>string</i>	optional suffix that terminates the jsonrpc frame.
	udp	<i>dictionary</i>	The UDP adapter configuration
	udp.server	<i>string</i>	UDP server hostname or IP address. A multicast address (224.0.0.0/4) enables the multicast mode (see below).
	udp.snd_port	<i>integer</i>	UDP server listening port
	udp.rcv_port	<i>integer</i>	UDP answer reception port (on client side)
	udp.bufsize	<i>integer</i>	the maximum expected size of the buffer received while waiting for a jsonrpc frame.
	udp.multicast_if	<i>string</i>	multicast mode only: IP address of the local interface used to send to and join the group (default: the kernel default interface).

Multicast

When `udp.server` is a multicast group address (224.0.0.0/4), the open action sends the requests to the group and joins it on the reception socket.

The response scheme is decided by the *server*, not by *testium*; no client configuration is needed, both are received transparently on `udp.rcv_port`:

- the server answers in unicast to the source address of the request (the usual case): received because the socket is bound to `rcv_port`,
- the server answers to the multicast group itself (discovery-style protocols): received because open joined the group. The server must then address the group on the client reception port.

Use `udp.multicast_if` to pin the exchanges to a given local interface (e.g. 127.0.0.1 for a same-host test bench); otherwise the kernel routing decides which interface carries the multicast traffic — on a multi-homed host the request may leave through the wrong network.

Warning

The UDP socket is created once by the open action and shared, through `rcv_port`, by every later query/receive item. `multicast_if` only acts at socket creation: set it on the item holding the open action. On any other item it is ignored (a warning is logged).

open action

The open jsonrpc action is only used with the *UDP adapter* but is mandatory before any query action.

No parameter is required.

close action

The close jsonrpc action is only used with the *UDP adapter* but is mandatory after JSONRPC transfers are finished.

No parameter is required.

query action

The query jsonrpc action has the following attributes:

- method: JSONRPC method to be called,
- params: JSONRPC param (must conform to the version defined above), by default it is an empty list.
- id: JSONRPC id. If not defined or starts with rand, it is chosen randomly. Otherwise it must be an integer value,
- timeout: reception timeout in seconds. It is a floating point number. It is by default the jsonrpc timeout.
- no_wait: Optional boolean. False by default. This attribute defines if the reception is performed in this step (reception can be done separately, in the receive action described below),

receive action

The receive jsonrpc action has the following attributes:

- id: JSONRPC id as an integer value,
- timeout: reception timeout in seconds. It is a floating point number, It is by default the jsonrpc timeout.

4.3.15 let test item

This element is of the following form:

Listing 4.26: let test item usage example

```
- let:
  name: Let Item
  values:
    - key1: value1
    - key2: value2
    - key3: <| $(variable)[$(loop_index)] |>
```

The let element is used to set values in the global dictionary.

Attributes

- The values list gives the {<key>, <value>} pairs to set in the global dictionary,

4.3.16 loop test items

This element is of the following form:

Listing 4.27: loop test item usage example

```
- loop:
  name: Cycle Temperature
  iterator: 10
  steps:
    - unittest:
      test_file: test_prod_rio6_8093.py
    - py_func:
      name: function test item
      file: scriptTestFile.py
      func_name: funcToBeExecuted
      param:
        - $(loop_param)
  exit_condition:
    file: script_name.py
    func_name: methodName
```

The loop element executes repeatedly the steps sequence of items.

Iteration is controlled by the iterator attribute, described later in this chapter.

Attributes

Below are described loop test item specific attributes.

- Iterator: giving the number of loop iteration (see dedicated chapter below).
- steps: describes the sequence executed at each cycle; it is a list of any of the testium test items.
- exit_condition: defines when to exit the loop. If the condition evaluates to True the loop continues; otherwise it stops. exit_condition attributes are:
 - time: the loop stops after the time (in minutes) is elapsed (optional)
 - value: the loop stops when the content of the value attribute is True (optional)
 - file: the script file that contains a function executed at each iteration. Only python script format is supported (optional if another exit_condition attribute is defined)
 - func_name: the function to execute on each loop when the file attribute is defined. The function referenced by the func_name attribute must have two parameters: the current loop iterator value and the report, even if they are not used. This attribute is mandatory if the file attribute is defined.
 - eval: optional parameter allowing post-processing of the function result. It is a python evaluable string in which the \$(result) keyword is replaced by the actual function call result (see example below).

Listing 4.28: loop exit condition

```
- loop:
  ...
  exit_condition:
    file: script_name.py
    func_name: methodName
    eval: $(result) < 2
```

Iterator

The iterator attribute can be of the following types:

- An integer giving the number of iterations,
- A list. The number of elements of the list gives the loop number, and the list member are the consecutive loop parameters,
- Undefined. Then cycle loops until the exit condition is reached.

Loop variables

The following loop variables are automatically defined:

- `$(loop_param)`: parameter of the loop. It contains the iterator value.
- `$(loop_index)`: index of the loop, starting with 0 and incremented at each cycle.
- `$(loop_index_inverse)`: index counting down, starting from the iteration count minus 1 and decremented at each cycle.
- `$(loop_count)`: loop total iteration number. If the number of loops is undefined its value is the python `inf`.

When these variables are found in a parameter, an attribute, etc, testium searches the test hierarchy upward for the enclosing loop and substitutes that loop's value.

If more than one loop exists in the test item hierarchy, the lowest level loop iterator is used.

4.3.17 lua_func test item

The `lua_func` test item is used to execute custom lua 5.4 scripts with the given input parameters.

The `lua_func` test item is of the form:

Listing 4.29: `lua_func` Lua function example

```
local module = {}

function module.dummy_func(param1, param2, param3, param4)
    ...
    return 10
end

return module
```

Listing 4.30: corresponding `lua_func` tum extract

```
- lua_func:
  name: lua function test item
  file: script_file.lua
  func_name: dummy_func
  param:
    - 123
    - 0.123
    - True
    - $(global_dict_key)
  expected_result: 10
```

Attributes

Besides the common test item attributes, the `lua_func` item has specific attributes; `file` and `func_name` are mandatory.

- `file`: the script file name that contains the function to be executed. Only Lua script format is supported.
- `func_name`: The function name to be executed.
- `param`: This is a list of parameters that are passed to the function in the order they are presented in the script. These parameters are not mandatory and depend on the function prototype.
- `context_id`: Optional. When set, all `lua_func` items sharing the same `context_id` value run inside the same persistent Lua subprocess for the duration of the test. See [lua_func context](#) for details.

Listing 4.31: `lua_func` test item example of usage

```
- lua_func:
  name: activity
  file: script_name.lua
  func_name: methodName
  param:
    - $(my_param)
```

The result of the function (after optional post-processing) is stored in the global variable named `lfn_<item_name>` (See [global variables](#) for more detail on how to access to global variables from test items and scripts).

In the example above, the global variable `$(lfn_activity)` would be created at the end of the item execution. It would contain the resulting value of the `methodName` Lua function.

The `lua_func` result is always PASS, unless the called function raises an error or the `expected_result` attribute is used.

If the called function returns several values, they are kept as a list in the result; a single return value stays a scalar.

Sharing state between `lua_func` calls

Each `lua_func` item without a `context_id` runs in a dedicated subprocess that is started and stopped around the call. Module-level variables are not preserved between two such items.

Inside a `lua_func` script, the `tm` module exposes `tm.setgd` and `tm.gd` to read and write the testium global dictionary of the test process. Values stored this way are accessible from any subsequent test item without requiring a shared subprocess.

Listing 4.32: sharing a value via the global dictionary

```
local tm = require("tm")
local module = {}

function module.produce(val)
  tm.setgd("my_shared_value", val)
  return val
end

function module.consume()
  return tm.gd("my_shared_value")
end

return module
```

When `context_id` is set, all `lua_func` items that share the same identifier reuse the same persistent subprocess. This allows Lua-side state (upvalues, module cache) to be retained across calls beyond what `tm.setgd` persists.

Listing 4.33: `lua_func` items sharing a persistent subprocess

```
- lua_func:
  name: produce value
  file: my_script.lua
  func_name: produce
  context_id: my_context
  param:
    - hello

- lua_func:
  name: consume value
  file: my_script.lua
  func_name: consume
  context_id: my_context
  expected_result: hello
```

The shared subprocess is automatically stopped at the end of the test run.

Lua Interpreter environment setup

Some global variables have an impact on the `lua_func` test item behavior:

- `lua_bin`: This optional global variable can be used to define the lua executable path. If not defined, the lua interpreter is looked up in the default system locations (`PATH`).
- `lua_env`: This global variable can be used to define environment variables for the lua script execution environment. Only `PATH`, `LUA_PATH`, and `LUA_CPATH` are supported.

Listing 4.34: example of configuration file: `param.yaml`

```
[...]
lua_env:
  PATH: "/my/path/"
  LUA_PATH: "/my/lua/modules/?.lua;;"
  LUA_CPATH: "/my/lua/modules/?.so;;"
[...]
```

4.3.18 parallel test item

This element is of the following form:

Listing 4.35: parallel test item usage example

```
- parallel:
  name: My parallel block
  sync: all
  branches:
    - name: Branch A
      steps:
        - py_func:
            name: Long operation
            file: long_op.py
            func_name: do_work
    - name: Branch B
```

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```
wait_for:
  condition: <| "$(ready_flag)" == "True" |>
  timeout: 30
steps:
  - let:
      name: Mark done
      values:
        - branch_b_done: true
```

The parallel element runs several sequences of items concurrently. Each inner sequence is called a *branch* and runs in its own thread. The parent test item waits for branches to finish according to the sync policy.

Attributes

- **branches:** required. A list of branches to execute concurrently. Each branch has a name and a steps list (same structure as a group item). It can also declare a `wait_for` precondition (see below).
- **sync:** optional, defaults to `all`.
 - `all`: the parallel item completes when every branch has finished. The result is PASS if no branch returned FAIL (skipped or disabled branches are ignored, like in group); otherwise FAIL.
 - `any`: the parallel item completes as soon as the *first* branch finishes. The remaining branches are stopped (their next test items are not executed). The result is PASS if at least one branch succeeded.
- **no_fail:** optional. When true, a FAIL result is forced to PASS for the parallel item itself (same semantics as for any test item). Branches keep their own result.

Branch attributes

Each entry of branches is a dict with the following attributes:

- **name:** required. The branch name. Used in reports and as a prefix in the live log output (each line printed by the branch is prefixed with the branch name in square brackets, e.g. [Branch A], so concurrent branches stay readable).
- **steps:** required. The list of test items executed sequentially inside the branch.
- **wait_for:** optional. Forces the branch to wait until a condition is met before running its steps. If the timeout elapses, the branch returns FAIL (the steps are not run). Sub-attributes:
 - **condition:** a testium expression evaluated repeatedly (every 100 ms) until it returns True.
 - **timeout:** maximum wait, in seconds. Defaults to 30.

Reporting

Each branch produces its own row in the SQLite report (with type `Parallel branch`), in addition to the parent `Parallel` row. The log column of each row contains only the output emitted from that branch's thread, so logs are never mixed between concurrent branches.

In the live (terminal / GUI) output, lines emitted from a branch are prefixed with the branch name in square brackets (e.g. [Branch A]). The prefix is not stored in the SQLite log column.

Notes

- A sleep item inside a branch is interruptible: if another branch finishes first under sync: any, slow sleep items are aborted within ~50 ms.
- A py_func or console item inside a branch is **not** interruptible: a sync: any stop will only take effect after the current item returns. The branch will then skip its remaining steps.
- When a user disables a branch in the GUI tree, the branch returns SKIP instantly without affecting the others (it does *not* count as the first finished branch under sync: any).

4.3.19 plot test item

This test item is used to display runtime values of test variables, or any changing value, in a separate external window.

The plot window is defined using the plot test item:

Listing 4.36: plot test item usage example

```
- plot:
  name: test name in GUI
  plot_name: plot identifier
  steps:
    - open:
    - add:
    ...
```

Attributes

In addition to common test items attributes, the plot test item has specific attributes:

- plot_name: plot window instance name.
- steps: a sequence of actions to be applied to the plot window. More than one action can be executed in a plot item.

The plot test item can accept the actions described in further sections.

All the following actions support the name attribute. The name is concatenated with the action type in the *testium* GUI, and recalled in the test log and reports.

open action

This action initializes and opens the plot window with the corresponding attributes as defined below.

This action accepts one optional log_path parameter defining the path where the plot line values are stored in CSV format.

Listing 4.37: plot open action

```
- plot:
  name: Open the plot window
  plot_name: plot identifier
  steps:
    - open:
      name: open the plot
      log_path: $(test_directory)/tmp
```

close action

The close action closes the plot window and removes it from the managed instances of *testium*.

This action does not have mandatory parameters. However, close optional action parameters are:

- `wait_dialog_exit`: Boolean value. If set to `True`, the window is kept open until the user closes it manually.
- `timeout`: Value expressed in seconds. It is active if the `wait_dialog_exit` is set to `True`. If this parameter is defined, and if not closed manually, the dialog window is kept open until the timeout elapses.

Listing 4.38: plot close action

```
- plot:
  name: Closes the plot
  plot_name: plot identifier
  steps:
    - close:
      wait_dialog_exit: True
      timeout: 600
```

Note

When the close action is entered, the periodic plots are stopped.

add action

The add action is used to add a single data point to the plot window.

Listing 4.39: plot add action

```
- plot:
  name: Add to the plot
  plot_name: plot identifier
  steps:
    - add:
      name: add value 1 & 2
      value1: ${loop_index}
      value2: ${loop_index}+2
```

The parameter of the add action is a dictionary of (*key*, *values*) pairs where the *key* is the plot line name and *value* is the numeric value to add to the plot line.

If the *value* is a string rather than a number, it is evaluated as a python expression.

periodic action

This action calls a python function periodically and uses its result to update the plot.

periodic plots are updated automatically and don't require further steps in a test sequence, once executed.

periodic action parameters are:

- `period`: period of the automatic value update.
- `file`: python file containing the function to call.

- `func_name`: the name of the python function to be periodically called.
- `eval`: optional parameter allowing post-processing of the function result.

The `eval` parameter of the periodic action is a python evaluable string in which the `$(result)` keyword is replaced by the actual function call result.

The result of the action must be a dictionary of (*key*, *values*) pairs where the *key* is the plot line name and *value* is the numeric value to add to the plot line.

Listing 4.40: plot periodic action

```
- plot:
  name: Add periodic to the plot
  plot_name: plot identifier
  steps:
    - periodic:
      period: 1
      file: $(test_path)$(psep)plot.py
      func_name: random_value
      eval: '{"periodic": $(result)}'
```

last_value action

The `last_value` action stores the last values added to the plot (periodically or not) in the global variables entries.

`last_value` action parameters are:

- `name`: Optional parameter giving the list of plot lines (measurements) to return. If it is not defined, all the plot lines are returned.

Listing 4.41: plot last_value action

```
- plot:
  name: Plot measure_1 value
  plot_name: plot identifier
  steps:
    - last_value:
      name: [measure_1]
```

The result of the action is stored in the global variable named `plv_<item_name>` in the example above, it would be `$(plv_Plot measure_1 value)`. See [global variables](#) for more detail on how to access to global variables from test items and scripts.

export action

The export action saves the plot window data in various formats to the filesystem.

Listing 4.42: plot export action

```
- plot:
  name: Plot export
  plot_name: plot identifier
  steps:
    - export: $(my_custom_path)/plot_export.pdf
    - export: $(my_custom_path)/plot_export.csv
```

At the time of writing of this documentation, `.pdf` and `.csv` files are supported.

4.3.20 py_func test item

The `py_func` test item is used to execute custom python scripts with the given input parameters.

There are two modes for executing a `py_func` item. The *class* mode and the *function* mode.

class py_func item

This is the normal way of calling some custom python code.

A class must be defined and derived from `FunctionItem` from the `py_func.tm` module.

From this class it is possible to define some custom reported values with the following API

- `reportValue(key, value)`: This `FunctionItem` method adds a value to the report,
- `reportedValues()`: This `FunctionItem` method returns the values reported so far.

Listing 4.43: `py_func` test item implementation example

```
import py_func.tm as tm

class TestItemPyFunc(tm.FunctionItem):

    def exec(self, param1, param2, param3, param4):
        ...
        self.reportValue('my_reported_value', reported_value)
        print(self.reportedValues())
        return 10
```

The `exec` method of the `FunctionItem` derived class is executed while running the `py_func` test item.

Listing 4.44: class `py_func` test item usage

```
- py_func:
  name: function test item
  file: scriptTestFile.py
  func_name: TestItemPyFunc
  param:
    - 123
    - 0.123
    - True
    - $(global_dict_key)
  expected_result: 10
```

legacy py_func

The legacy `py_func` test item is of the form:

Listing 4.45: legacy `py_func` python function example

```
def dummy_func(param1, param2, param3, param4):
    ...
    return 10
```

There is no possibility to access the report features in that mode.

Listing 4.46: corresponding py_func tum extract

```
- py_func:
  name: function test item
  file: scriptTestFile.py
  func_name: funcToBeExecuted
  param:
    - 123
    - 0.123
    - True
    - $(global_dict_key)
  expected_result: 10
```

Attributes

Besides the common test item attributes, the `py_func` item has specific attributes; `file` and `func_name` are mandatory.

- `file`: the script file name that contains the function to be executed. Only python script format is supported.
- `func_name`: The function name to be executed.
- `param`: This is a list of parameters that are passed to the function in the order they are presented in the script. These parameters are not mandatory and depend on the function prototype.
- `context_id`: Optional. When set, all `py_func` items sharing the same `context_id` value run inside the same persistent Python subprocess for the duration of the test. See [py_func context](#) for details.

Listing 4.47: py_func test item example of usage

```
- py_func:
  file: script_name.py
  func_name: methodName
  param:
    - $(my_param)
```

The result of the function (after optional post-processing) is stored in the global variable named `pfn_<item_name>` (See [global variables](#) for more detail on how to access to global variables from test items and scripts).

In the example above, the global variable `$(pfn_function test item)` would be created at the end of the item execution. It would contain the resulting value of the `funcToBeExecuted` python function.

The `py_func` result is always PASS, unless the called function raises an exception or the `expected_result` attribute is used.

Sharing state between py_func calls

Each `py_func` item without a `context_id` runs in a dedicated subprocess that is started and stopped around the call. State cannot be shared between two such items using module-level variables.

Inside a `py_func` script, `tm.setgd` and `tm.gd` read and write the testium global dictionary. Values stored this way are accessible from any subsequent test item, including other `py_func` items, without requiring a shared subprocess.

Listing 4.48: sharing a value via the global dictionary

```
import py_func.tm as tm

def produce(val):
    tm.setgd("my_shared_value", val)
    return val

def consume():
    return tm.gd("my_shared_value", None)
```

When `context_id` is set, all `py_func` items that share the same identifier reuse the same persistent subprocess. This allows sharing any Python object across calls — including objects that cannot be transmitted to other processes.

Listing 4.49: sharing an object via `context_id`

```
import py_func.tm as tm

def open_connection():
    tm.setgd("conn", MyConnection())
    return "ok"

def use_connection():
    conn = tm.gd("conn")
    return conn.status()
```

Listing 4.50: `py_func` items sharing a persistent subprocess

```
- py_func:
  name: open connection
  file: my_script.py
  func_name: open_connection
  context_id: my_context
  expected_result: ok

- py_func:
  name: use connection
  file: my_script.py
  func_name: use_connection
  context_id: my_context
  expected_result: open
```

The shared subprocess is automatically stopped at the end of the test run.

Python Interpreter environment setup

Some global variables have an impact on the `py_func` test item behavior:

- `python_bin`: This optional global variable can be used to define the python executable path. If not defined, the python interpreter is looked up in the default system locations.
- `python_env`: This global variable can be used to define environment variables for the python script execution environment. Only `PATH` and `PYTHONPATH` are supported.

Listing 4.51: example of configuration file: `param.yaml`

```
[ ... ]
```

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```
python_env:
  PATH: "/my/path/"
  PYTHONPATH: "/my/python/modules/"
  [ ... ]
```

4.3.21 pytest test item

The pytest test item runs a [pytest](#)⁸ test file and turns every collected test into a child item, each with its own PASS / FAIL / SKIP result, duration and failure message. It is the pytest counterpart of the unittest test item.

The tum file prototype is as follows:

Listing 4.52: pytest test item usage example

```
- pytest:
  name: pytest test item
  test_file: test_device.py
  test_method:
    - test_boot
    - test_version
```

Attributes

Beside common test items attributes, the pytest test item has specific attributes:

- `test_file`: the name (and optionally the path) of the pytest file to run.
- `test_method`: optional list of test function names. When present, only the matching tests are included in the test tree (the name is matched against the function part of each pytest *node id*, the parametrisation suffix being ignored). Otherwise every collected test in the file is run.

Host execution

Unlike unittest (which runs in *testium*'s own interpreter), the pytest item runs pytest in a **subprocess on the host interpreter** — the same one used by `py_func` / `lua_func` (overridable with the `python_bin` global variable). As a consequence:

- pytest (and the test's own dependencies) must be installed on that host interpreter — e.g. `pip install pytest`. It is not bundled with *testium*.
- the tests run isolated from *testium*'s in-process API; they are meant to be self-contained (they exercise the device/software under test, not the *testium* tree).

The item is invoked with a controlled pytest configuration (`--capture=no`, user adopts neutralised, cache disabled); a small built-in pytest plugin reports the collected tests and their results back to *testium*.

4.3.22 report test item

This test item exports report files **during** the test run, from the data collected so far — useful to snapshot partial results in a long campaign.

It requires a report element at the root of the main test file (see the [Reports chapter](#)).

⁸ <https://docs.pytest.org>

Listing 4.53: report test item usage example

```
- report:
  name: Intermediate report
  export:
    - junit:
      path: $(home)/reports
      file_name: intermediate.xml
      pattern:
        - Unittest%
    - text:
      path: $(home)/reports
      file_name: report-key-1.txt
      key:
        - report-key-1
```

Attributes

Besides the *common attributes*, the report test item has one specific attribute:

- **export**: required. One export entry or a list of them, with exactly the same syntax and attributes (path, file_name, pattern, key, cmd) as the root-level export — see *export attributes*. All formats are available: *built-ins*, the *command export* and installed *plugins*.

Notes:

- a `sqlite` entry is ignored here: the database storage is configured once, in the root report element;
- exports run with `no_header` set — the produced files carry the test rows without the run header (the run is not finished, so the global result and duration do not exist yet).

4.3.23 run test item

This test item executes a new instance of testium with the specified .tum file.

- In **batch mode** (-b): the sub-instance is started with -b.
- In **GUI mode**: the sub-instance opens its own window with -r (run and close).
- `batch: true` forces the sub-instance to run headless (-b) even in the GUI.

A sub-instance started with -b is **captured**: its output is streamed into this test's log and report, and kept as the item's result value, so it can be stored with `store_result` and post-processed (`expected_result`, `process_result`, or a `py_func` reading the global variable).

The item result is **PASS** if the sub-instance launched and ran to completion, regardless of whether the sub-tests passed or failed. It is **FAIL** if the file could not be found, the sub-instance could not be launched, or the time window was not reached (see `start_time` / `end_time`).

Listing 4.54: run test item usage example

```
- run:
  name: Execute TUM
  tum: example_cycle.tum
  log_file: $(home)/reports/test.log
  report_file: $(home)/reports/test.rep
```

Attributes

run test item has the following specific attributes:

- **tum**: mandatory, the path of the file to execute. Can be relative to the current execution folder.
- **param_file** (optional): the path of the parameter file to use; otherwise the default parameter file is used.
- **batch** (optional): **true** to run the sub-instance headless (-b) and capture its output even in GUI mode (see above).
- **log_file** (optional): the path of the log file. In GUI mode, if not provided, a file is created with a timestamp next to the .tum file. Not used in batch mode (the output is captured instead).
- **report_file** (optional): the path of the report file to create.
- **start_time** (optional): earliest time to execute the sub-instance, in HH:MM format.
- **end_time** (optional): latest time for execution within a time frame, in HH:MM format.
- **wait_for_exec** (optional): **true** to wait until the time window defined by **start_time** and **end_time** is reached before running. Requires both **start_time** and **end_time**.

4.3.24 sleep test item

sleep test item has the following description format

Listing 4.55: sleep test item example usage

```
- sleep:
  name: sleep test item
  timeout: 10
  dialog: True
```

Attributes

- **timeout**: sleep duration in seconds, or a relative duration string such as “2d 5h 31m 3s” (2 days, 5 hours, 31 minutes and 3 seconds).
- **dialog**: If set to **True**, a window showing the remaining time to wait is displayed (optional parameter set to **False** by default)

4.3.25 unittest test item

The unittest test item runs a test script written with the `unittest` module from the python standard library.

The tum file prototype is as follows:

Listing 4.56: unittest test item usage example

```
- unittest:
  name: unitTest test item
  test_file: unitTestScript.py
  test_method:
    - test_1
    - test_2
```

Attributes

Besides the common test item attributes, the unittest test item has specific attributes; `test_file` is mandatory.

- `test_file`: it is the name (and optionally the path) of the unittest file to be processed.
- `test_method`: it is an optional unittest test sub-item. If one or more elements are present, the unittest python script file is parsed and only the corresponding methods are included in the test tree. Otherwise, all the test methods are included in the test tree.

Access to global variables entries

unittest file tests instances have access to the testium global variables by using the *helper's library*.

Report value from unittest

Values can be added to the test report from a unittest test at runtime.

Listing 4.57: example of unittest test item python function

```
from unittest import (TestCase)

class DummyTests(TestCase):
    def test_01_report(self):
        self.reported_values['key reported'] = 'value_reported'
```

Console use example with unittest item

Here is an example how to use the console module from python unittest.

Listing 4.58: example of a *testium* console usage from a unittest python function

```
from unittest import (TestCase)
import console

class DummyTests(TestCase):
    @classmethod
    def setUpClass(cls):
        cls.consA0 = console.TelnetConsole('cons name', '192.168.98.123',
→7001)
        cls.consA0.open()
        cls.promptA0 = 'test-computer>'

    def test_01_console(self):
        self.consA0.write('config')
        self.assertEqual(self.consA0.read_until(self.promptA0, 10), 0)
        self.consA0.write('lsusb && echo "Done."\n')
        status, read_data = self.consA0.read_until('Done.',
                                                    10, return_data=True)
        self.assertEqual(status, 0)
        if read_data.find('ID 04f2:b684 Chicony Electronics Co.') != -1:
            index=0
```

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```
@classmethod
def tearDownClass(cls):
    cls.consA0.close()
```

4.4 Includes

It is possible to include one TUM file in another by using the `!include` tag before the file name.

This feature is a testium specific implementation and is not part of the YAML language, although it is based on the tagging feature of the language and the customization possibility offered by the python pyYaml package.

Here is a basic example of file inclusion:

Listing 4.59: included_file.tum

```
- test_item:
  name: test_2
- test_item:
  name: test_3
```

Listing 4.60: main.tum

```
#include with the sub-sequence reference mechanism
sequence: &included_sequence
  !include included_file.tum

main:
  name: Test example
  steps:
    - test_item1:
      name: test_1
    - *included_sequence

#include can also be inserted directly within the steps list
  - !include included_file.tum
```

4.5 Templates

testium embeds the [jinja2](https://jinja.palletsprojects.com)⁹ template engine. It allows extensive customization of test files and makes test scripts reusable.

4.5.1 In the main test file

The *testium* main test files are always passed through the jinja template engine.

The parameters passed to jinja are all the variables contained into the *configuration files* plus the *built-in values*.

⁹ <https://jinja.palletsprojects.com>

4.5.2 In !include directive

In addition to basic inclusion, !include accepts parameters. These parameters replace the corresponding {{ keyword }} placeholders in the included file.

See examples below.

Listing 4.61: including a template

```
main:
  name: Test example
  steps:
    - test_item1:
      name: test_1

    #include can also be inserted directly within the steps list
    - !include
      file: included_template_file.tum
      inclusion_parameter_1: param1
      inclusion_parameter_2: param2
```

Listing 4.62: included template

```
- test_item:
  name: {{ inclusion_parameter_1 }}
- {{ inclusion_parameter_2 }}:
  name: test_3
# The following construction is not allowed and will fail to load:
- test_item:
  name: {{ $(inclusion)_parameter_3 }}
```

4.6 Reports

During a test run, testium records one row per executed test item — name, type, result, message, duration, reported values — into a SQLite database, together with a header describing the run (test file, date, global result, ...). A report **export** turns that database into an output file: JUnit XML, HTML, JSON, plain text, the raw database itself, or any custom format.

This chapter describes:

- how to enable the report and declare exports (*declaring exports*),
- the attributes accepted by every export entry (*export attributes*),
- the built-in formats and the special role of sqlite (*built-in formats*),
- post-processing with an external program (*command export*),
- adding custom formats as pip plugins (*exporter plugins*),
- the report database schema (*database schema*).

4.6.1 Declaring exports

The report element at the root of the main .tum file enables the recording and lists the exports. The export key takes one entry or a list of entries; each entry uses the format name as its key:

Listing 4.63: report declaration — multiple exports

```

report:
  enabled: True
  log_stored: True
  export:
    - sqlite:
        path: $(home)/reports
        file_name: $(test_name).db
    - junit:
        path: $(home)/reports
        file_name: $(test_name).xml
    - html:
        path: $(home)/reports
        file_name: $(test_name).html

```

Table 4.4: report attributes

Attribute	default value	Description
enabled	True	Enables report recording.
log_stored	False	When True, the stdout of each test item is captured during the run, so that exports (html, json) can include the log of each item.
export	/	One export entry or a list of them. Each entry's key is the format name (see Built-in formats , command export — external tool , Custom export formats (plugins)).

When the exports run:

- **at the end of the run** — each entry of the export list above produces one output file, in declaration order. This is the normal case.
- **during the run** — the [report test item](#) placed in the test sequence runs its own export list against the data collected so far. Useful to snapshot partial results in a long campaign.
- `sqlite` is the exception: it is not converted at the end but written live during the run (see [Built-in formats](#)).

In [batch mode](#), the `-p`, `-t` and `-n` command-line options (see `-p`) **replace** the export list of the test file with a single export built from the given path, type and patterns.

4.6.2 Export attributes

Every export entry accepts the following sub-attributes:

Table 4.5: export attributes

Attribute	default value	Description
path	\$(report_path)	Output directory.
file_name	/	Output file name.
pattern	/	One or more SQL LIKE patterns applied on the test item name.
key	/	One or more SQL LIKE patterns applied on the test item key attribute.
cmd	/	command export only: the external command line to run (see command export — external tool).

path and file_name are joined to build the output file; you can also put the full path in just one of the two attributes. All attributes accept `$(...)` [global variable expansions](#), resolved at export time — file_name: `$(test_name).xml` produces one report file per test file, for example. If the output file already exists, it is renamed to `<name>-<N>.saved` first, never overwritten.

pattern and key select which test items appear in the output. Without them, every executed item is exported. Both use the SQL LIKE syntax — % matches any text, _ one character — and accept a single string or a list; an item is kept when it matches **any** of the given patterns.

- pattern matches on the item name;
- key matches on the item key attribute, one of the [common test item attributes](#). The usual scheme is to tag related items with the same key in the .tum and filter the report on it:

Listing 4.64: selecting items with key

```
steps:
  - console:
      name: power on
      key: setup
      ...

report:
  export:
    - junit:
        file_name: setup_only.xml
        key: setup
```

4.6.3 Built-in formats

- sqlite — the report database itself (see below).
- text — indented text dump of the test tree.
- json — full report as JSON: `{"header": {...}, "tests": [...]}`.
- junit — JUnit XML, for CI systems (requires the junit_xml Python package).
- html — single HTML page with header, results table and per-item logs (requires lxml).

sqlite has a special role: it is not a conversion but the storage layer itself. With a sqlite entry, the internal database is written **live** to the given file during the run; the file remains afterwards for any external analysis (schema in [Report database schema](#)). Without it, the database only lives in memory and disappears once the other exports have run. Declaring several sqlite entries is pointless — only the last path is used.

The `html` and `json` exports can embed the captured stdout of each test item; this requires `log_stored: True` in the report block (see [Declaring exports](#)).

If a format is unknown, or an optional dependency is missing, the export is skipped with an `[report] Export skipped: ...` line on stdout and the test run **continues**. The line lists the available formats: built-ins, `command`, plus every installed plugin.

4.6.4 command export — external tool

The `command` export post-processes the report with any external program. The report database is copied to a temporary SQLite file and the command runs on the host system (even from the Flatpak / AppImage sandboxes):

Listing 4.65: post-processing the report with an external tool

```
report:
  export:
    - command:
      cmd: mytool --input $(db) --output $(out)
      path: $(home)/reports
      file_name: $(test_name).pdf
```

Two placeholders, following the usual `$(...)` syntax, are reserved in `cmd` (they are resolved by the export itself, not from the global variables):

- `$(db)` — path of the temporary SQLite copy of the report (schema in [Report database schema](#));
- `$(out)` — the output path built from `path` / `file_name`.

`cmd` is split shell-style (quotes group words) but **not** run through a shell: no pipes, redirections or globbing — wrap them in a script if needed. Other `$(...)` [expansions](#) work as usual, so `$(python_bin) myscript.py $(db) $(out)` runs a script with the same interpreter as `py_func`. The command runs with the test directory as working directory, so relative paths behave as elsewhere in the `.tum` file. The temporary `$(db)` copy is deleted after the command returns — the command must not rely on it afterwards.

The command output (stdout) is forwarded to the test log. A command that cannot be started or exits with a non-zero code produces an `[report] Export skipped: ...` line (with its stderr); the test run continues.

4.6.5 Custom export formats (plugins)

A third-party Python package can register additional export formats via the `testium.exporters` setuptools entry point group. A step-by-step tutorial (package layout, install, use) is provided in `doc/exporter_tutorial.md` of the source repository; this section is the reference.

The plugin must be installed with the **host** Python interpreter — the one resolved as `python_bin` and used by `py_func`. A plain `pip` install with that interpreter is enough; no `testium` configuration change is needed. The format is then usable in any `.tum` by its declared name, in **every** install channel: source, wheel, PyInstaller, Flatpak and AppImage.

Execution model: any format name that is not a built-in is looked up on the host. `testium` copies the report database to a temporary SQLite file, spawns the host Python, scans the installed `testium.exporters` entry points there, loads the matching class and instantiates it with the database copy. Consequences:

- the plugin (and any dependency it imports) must be installed with `python_bin`'s `pip` — not inside the `testium` bundle, which is read-only in the frozen channels;

- the plugin cannot use testium internals — it only sees the SQLite database (*Report database schema*) and its constructor arguments;
- a crash in the plugin (or a missing plugin dependency) skips that export with an [report] Export skipped: ... line and the run continues.

The plugin declares its format name in its `pyproject.toml`:

Listing 4.66: registering an exporter via entry-points

```
[project.entry-points."testium.exporters"]
my_format = "my_pkg:MyExporter"
```

The testium_report helper

The recommended way to write an exporter is the `testium_report` helper module shipped with testium (the built-in formats are implemented with it). It is importable by the plugin with **no installation step**: the process loading the plugin class resolves it from the running testium, so its version always matches. Subclass `Exporter` and implement `export()`:

Listing 4.67: exporter built on the helper

```
from testium_report import Exporter

class MyExporter(Exporter):
    def export(self):
        # self.rows      : filtered items - .name .type .key .result
        #                  (.passed/.failed/.skipped), .message,
        #                  .duration_s, .level, .log, .data (decoded)
        # self.report     : .header dict, .rows(pats, keys), .tree()
        # self.out_path   : output path (previous file renamed .saved)
        # self.name / self.no_header : report name, inline-item call
        with open(self.out_path, "w") as f:
            for row in self.rows:
                f.write(f"{row.name}: {row.result}\n")
```

Report also accepts a database file path instead of a connection, which makes exporters easy to unit-test against a saved sqlite report. For the development environment (IDE, tests), `pip install testium` provides the same `testium_report` module as a regular top-level import; at execution under testium the shipped copy takes precedence.

Writing an exporter without the helper

The helper is optional. To run an export, testium only instantiates the plugin class with the arguments below — a class with this constructor can read the SQLite tables (*Report database schema*) directly, without `testium_report`:

Listing 4.68: minimal exporter class

```
class MyExporter:
    def __init__(self, name, con, path, pats, keys, no_header=False):
        # name      : str - report name
        # con       : sqlite3.Connection (read) - tables: header, tests
        # path      : str - output file path (variables already expanded)
        # pats      : list[str] - LIKE filters on test_name (may be empty)
        # keys      : list[str] - LIKE filters on report_key (may be empty)
        # no_header : bool - do not write the run header (set when the
        #                  export comes from an inline `report` test item)
        ... # do the work in __init__ and write to `path`
```

4.6.6 Report database schema

The SQLite database produced by the `sqlite export` — and passed to `command exports` and plugins as a temporary copy — holds two tables.

`header` — one (key TEXT, value TEXT) row per run property: `report_version`, `test_file`, `test_name`, `test_result`, `test_revision`, `testium_version`, `testrun_date`, `testrun_time`, `test_duration`.

`tests` — one row per executed test item, 12 columns:

Table 4.6: tests table columns

Column	Content
<code>timestamp_start</code>	Start timestamp; the table is ordered on it.
<code>test_id</code>	Unique item id.
<code>parent_id</code>	<code>test_id</code> of the enclosing item (tree structure).
<code>level</code>	Nesting depth.
<code>test_name</code>	Item name attribute.
<code>test_type</code>	Item type (Console, Sleep, ...).
<code>report_key</code>	Item key attribute (see <i>Items common attributes</i>).
<code>result</code>	PASS, FAIL or SKIP.
<code>message</code>	Result message.
<code>duration</code>	Item duration (0.1 ms units).
<code>log</code>	Captured stdout of the item, when <code>log_stored: True</code> .
<code>data</code>	JSON of the values reported by the item — e.g. <code>reportValue(...)</code> in a <i>py_func</i> .

4.7 Test outputs

A list of test result outputs is automatically updated by *testium*.

This is an entry of the global variables dataset whose key is `test_outputs`.

This `global_dict` member contains the log file path and, if configured, the report path as a list.

The user can add other logged files by updating this global variables entry.

4.8 Post execution

A post execution script can be run for example to copy the output files.

For that, a `post_execution` element can be defined in the `.tum` file.

If the test run succeeds, the `post_exec` function of the `file_name` module is run; otherwise `post_exec_fail` is run.

If the `post_execution` element is not defined, the `post_execution.py` file in the test directory is used by default if it exists.

Listing 4.69: custom post execution python file

```
post_execution:
  file_name: test_report_text.py
```

4.9 Sub-sequence references

It is possible to alias any part of the TUM description file (typically a sequence of steps to be executed) to be inserted within another sequence.

This feature uses the anchor/alias mechanism of the [YAML language](https://yaml.org/)¹⁰.

Here is an implementation example of a reference to a sub-sequence in a TUM file:

Listing 4.70: sub-sequences call

```
sequence: &temperature_step_sequence
- test_item:
  name: test_2
- test_item:
  name: test_3

main:
  name: Test example
  steps:
    - test_item1:
      name: test_1
    - *temperature_step_sequence
```

Note

The entry before the alias (sequence: in the example above) is required by YAML syntax. However, its value is not used by *testium* and can be anything.

4.10 Test documentation

It is possible to display explanatory text in the GUI.

The doc attribute of test items is used for that purpose and is displayed as a tooltip on the test row.

Listing 4.71: tests documentation

```
main:
  name: Test example
  steps:
    - unittest:
      name: unittest item
      doc: |
        The purpose of this unittest test item is to demonstrate
        its various features.
```

(continues on next page)

¹⁰ <https://yaml.org/>

(continued from previous page)

test_file: dummy/dummy.py
test_method: test_01_pass

See illustration in Figure 4.3.

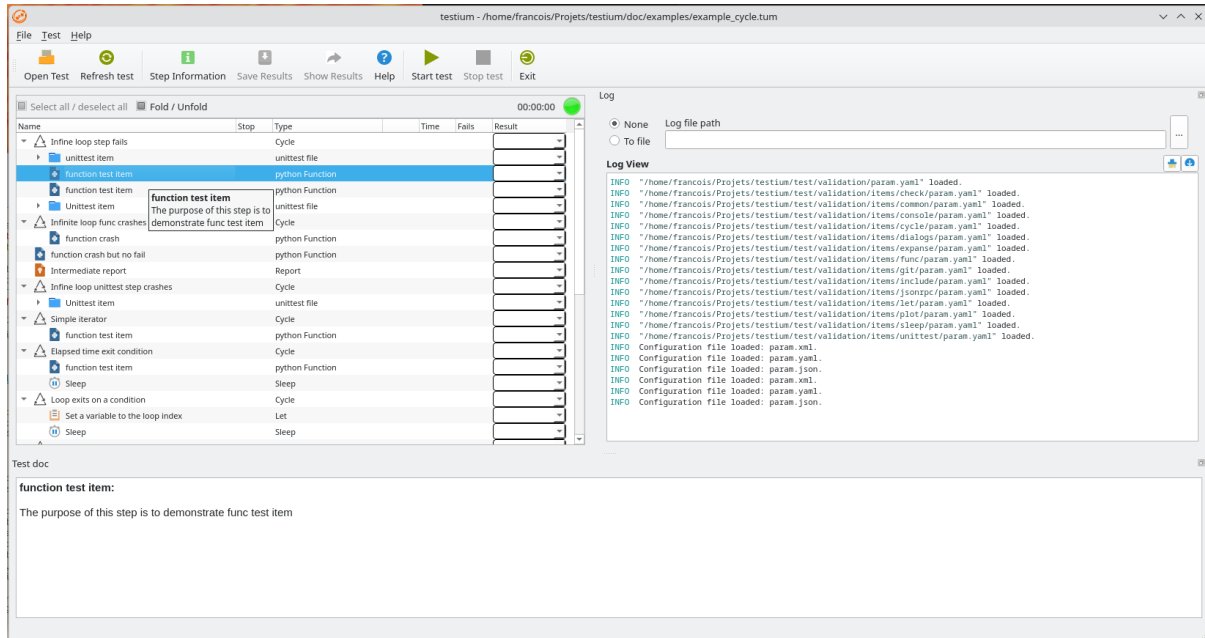


Fig. 4.3: Illustration of the doc attribute effect in the GUI.

4.10.1 Unittest

For unittest type test items, the python docstring of the test method is used as documentation.

PYTHON HELPER LIBRARY

A Python library of helper functions for Python modules called from testium `py_func` items. User scripts run inside the `py_func` subprocess and interact with testium through a JSON-RPC bridge — the `py_func.tm` module. They must **not** import `api.testium` or `interpreter.*` directly: those are main-process modules and may not even be reachable in a packaged build (PyInstaller, .deb).

To include the support of this library in a python script, the following line must be included in the script header:

Listing 5.1: testium helper library import

```
import py_func.tm as tm
```

5.1 Global variables helper functions

To manage values in the global variables dataset:

delgd(*name*)

Remove an entry from the testium global dictionary.

Parameters

name (*str*) – Name of the entry to remove.

gd(*name*, *default=None*)

Return a value from the testium global dictionary.

The value is accessible from any test item and from any `py_func` subprocess, regardless of the `context_id` used.

Parameters

- **name** (*str*) – Name of the entry to retrieve.
- **default** – Value returned when the key is absent. Defaults to None.

Returns

The stored value, or *default* if not found.

setgd(*name*, *value*)

Store a value in the testium global dictionary.

The stored value is accessible from any subsequent test item and from any `py_func` subprocess via `gd()`.

When `context_id` is used on the `py_func` item, any Python object (including those that cannot be transmitted to other processes) can be stored and shared between calls running in the same subprocess.

Parameters



- **name** (*str*) – Name of the entry to set.
- **value** – Value to store.

5.2 Plot helper functions

Add values to a running plot or read the last value from it:

add_plot_values(**params*)

last_plot_value(**params*)

Console and plot **lifecycle** management (`add_console`, `remove_console`, `console`, `add_plot`, `remove_plot`, `plot`) is performed by the console and plot test items themselves — not from user `py_func` scripts. Use those test items to open/close consoles and plots.

5.3 Other helper functions

OS(**params*)

get_main_dir(**params*)

init_timestamp(**params*)

text_mode(**params*)

timestamp(**params*)

timestamp_as_sec(**params*)

5.4 Debug mode

The `test_debug` global variable controls debug-only output. Read or write it via `tm.gd("test_debug")` / `tm.setgd("test_debug", True)`.